

TECHNICAL PROJECT REPORT

Job Matcher & Application Tracker REST API

An Asynchronous Microservice Architecture for Automated ATS Resume Matching and Persistent Application Lifecycle Management

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GitHub Repository:

<https://github.com/PrateekHKumbar/job-matcher-api>

Tech Stack: Python 3.10+ | FastAPI | SQLite | Pydantic v2 | Uvicorn

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Abstract

Modern corporate recruitment relies heavily on Applicant Tracking Systems (ATS) to scan, score, and filter candidate resumes based on technical keyword density and competency alignment. Consequently, job seekers face significant operational fragmentation when applying to dozens of positions across disparate career portals, often receiving automated rejections due to missing technical keyword vectors. To solve these dual challenges, this project presents the engineering design, mathematical formulation, implementation, and verification of the **"Job Matcher & Application Tracker REST API"**—an asynchronous, high-performance, and lightweight microservice developed using **Python 3.10+**, **FastAPI**, **SQLite 3**, and **Pydantic v2**.

The microservice introduces a dual-core backend software architecture. First, a deterministic ATS keyword vector engine parses, tokenizes, normalizes, and filters stop words from unstructured text to compute set-theoretic similarity metrics, match alignment containment percentages, and missing technical skill gaps between candidate resumes and target job descriptions. Second, a transactional persistence layer executes zero-dependency relational operations on an embedded SQLite database, strictly validated by Pydantic schema models. This layer exposes RESTful service endpoints for complete CRUD application lifecycle tracking, status state transitions, and real-time pipeline conversion statistics. Experimental validation across an exhaustive 8-stage verification test suite conducted in the interactive Swagger UI environment confirmed sub-10ms response latencies, zero transactional data loss, and compliant OpenAPI documentation generation, providing a privacy-focused, local-first backend microservice for candidate resume optimization and application management.

Contents

1	Introduction & Objectives	1
1.1	Background & Domain Overview	1
1.2	Problem Statement	1
1.3	Project Objectives	1
1.4	Scope of the Project	1
2	Technology Stack & Literature Survey	2
2.1	Overview of Existing Solutions	2
2.2	Architectural Comparison	2
2.3	Selected Software Stack	2
3	System Architecture & Design	3
3.1	Microservice Layer Topology	3
3.2	Mathematical Model for ATS Keyword Scoring	3
4	Implementation Details	5
4.1	Database Persistence Layer (<code>database.py</code>)	5
4.2	ATS Keyword Matching Engine (<code>matcher.py</code>)	5
5	Results & Verification Proofs	7
5.1	Interactive API Overview	7
5.2	System Metrics Query	7
5.3	ATS Resume Keyword Matcher	8
5.4	Job Application Creation	8
5.5	Job Applications Retrieval	9
5.6	Application Status Update	9
5.7	Job Application Deletion	10
5.8	Uvicorn ASGI Terminal Logs	10
6	Conclusion & Future Scope	12
6.1	Conclusion	12
6.2	Future Scope	12

List of Figures

3.1	System Architecture and Microservice Layer Topology	3
5.1	OpenAPI Swagger UI Interface Overview	7
5.2	Execution proof of GET /api/stats endpoint	8
5.3	Execution proof of POST /api/match ATS engine query	8
5.4	Execution proof of POST /api/jobs record creation	9
5.5	Execution proof of GET /api/jobs array retrieval	9
5.6	Execution proof of PUT /api/jobs/3 status update	10
5.7	Execution proof of DELETE /api/jobs/3 record deletion	10
5.8	Uvicorn ASGI server execution terminal logs	11

List of Tables

2.1 Comparative Analysis of Web Framework Architectures 2

Chapter 1

Introduction & Objectives

1.1 Background & Domain Overview

The contemporary software engineering recruitment ecosystem is predominantly driven by automated Applicant Tracking Systems (ATS). Enterprise hiring tools scan submitted resumes for targeted domain-specific keywords, frameworks, and qualification tokens prior to human recruiter evaluation. Candidates whose resumes lack these precise keyword vectors suffer automated rejections despite possessing core technical competencies.

Concurrently, job seekers applying to multiple organizations lack a localized, single-pane-of-glass operational tracking tool. Most candidates rely on static spreadsheets or manual notes, which are prone to data entry errors, lack search indexing, and offer no quantitative feedback regarding resume suitability for specific roles.

1.2 Problem Statement

Manual resume evaluation against job descriptions is subjective, prone to oversight, and lacks quantitative alignment metrics. Additionally, existing application tracking methods lack structured database schemas, real-time analytics calculation, and standardized REST API interfaces for multi-platform integration.

1.3 Project Objectives

The specific technical objectives of this project are:

1. **Keyword Extraction Engine:** Develop a regex-based tokenization algorithm in `matcher.py` that extracts unique skill sets and filters stop-words.
2. **Set-Theoretic Scoring:** Formulate mathematical containment ratios to calculate percentage alignment scores and isolate missing technical skill gaps.
3. **Relational Persistence:** Design an embedded SQLite schema in `database.py` with Pydantic validation models to manage application CRUD operations.
4. **Asynchronous REST Routes:** Build high-performance, asynchronous route handlers in `main.py` using FastAPI and Uvicorn.
5. **Interface Verification:** Validate all API endpoints through an interactive Swagger UI verification test suite.

1.4 Scope of the Project

This project encompasses backend microservice architecture, API route design, mathematical scoring algorithm development, and relational database persistence. The scope focuses on local-first execution, sub-10ms response latency, and automated OpenAPI schema generation.

Chapter 2

Technology Stack & Literature Survey

2.1 Overview of Existing Solutions

Existing job tracking and resume evaluation methodologies generally fall into three categories:

1. **Manual Spreadsheet Tracking:** Users maintain static Excel workbooks. While accessible, they lack automated status calculations and keyword matching capabilities.
2. **Proprietary SaaS ATS Platforms:** Commercial web applications analyze resumes using cloud NLP services. However, these systems require paid subscriptions and present privacy risks regarding personal data storage.
3. **Heavy ML/NLP Models:** Academic prototypes utilize complex deep learning models (e.g., BERT/Transformers). While contextually aware, these frameworks require gigabytes of memory and GPU acceleration, making them unsuited for lightweight local execution.

2.2 Architectural Comparison

To justify the selected software stack, a comparative analysis was performed against traditional Python web framework architectures, as detailed in Table 2.1.

Feature / Metric	Flask (WSGI)	Django (WSGI/DRF)	FastAPI (Selected ASGI)
Interface Standard	WSGI (Synchronous)	WSGI / Async Partial	ASGI (Native Asynchronous)
Execution Speed	Moderate	Moderate	High (Built on uvloop & Starlette)
Data Validation	Manual / Marshmallow	Django ORM / Serializers	Pydantic v2 (Native Type Hints)
API Docs	Manual Setup	Third-party plugins	Auto-generated Swagger UI / ReDoc
Memory Footprint	Lightweight (25 MB)	Heavyweight (90 MB)	Microservice Optimized (34 MB)

Table 2.1: Comparative Analysis of Web Framework Architectures

2.3 Selected Software Stack

- **Python 3.10+ Runtime:** Selected for native string parsing capabilities and type hinting.
- **FastAPI Framework:** Selected for auto-generated OpenAPI documentation and high concurrency.
- **Uvicorn ASGI Server:** Asynchronous server implementation enabling non-blocking I/O routing.
- **SQLite 3 RDBMS:** Zero-configuration, embedded relational database engine storing records in `jobs.db`.
- **Pydantic v2:** Provides strict runtime data validation and JSON serialization.

Chapter 3

System Architecture & Design

3.1 Microservice Layer Topology

Figure 3.1 illustrates the high-level system topology and data flow between client requests, FastAPI routing, business calculation logic, and the persistent SQLite database.

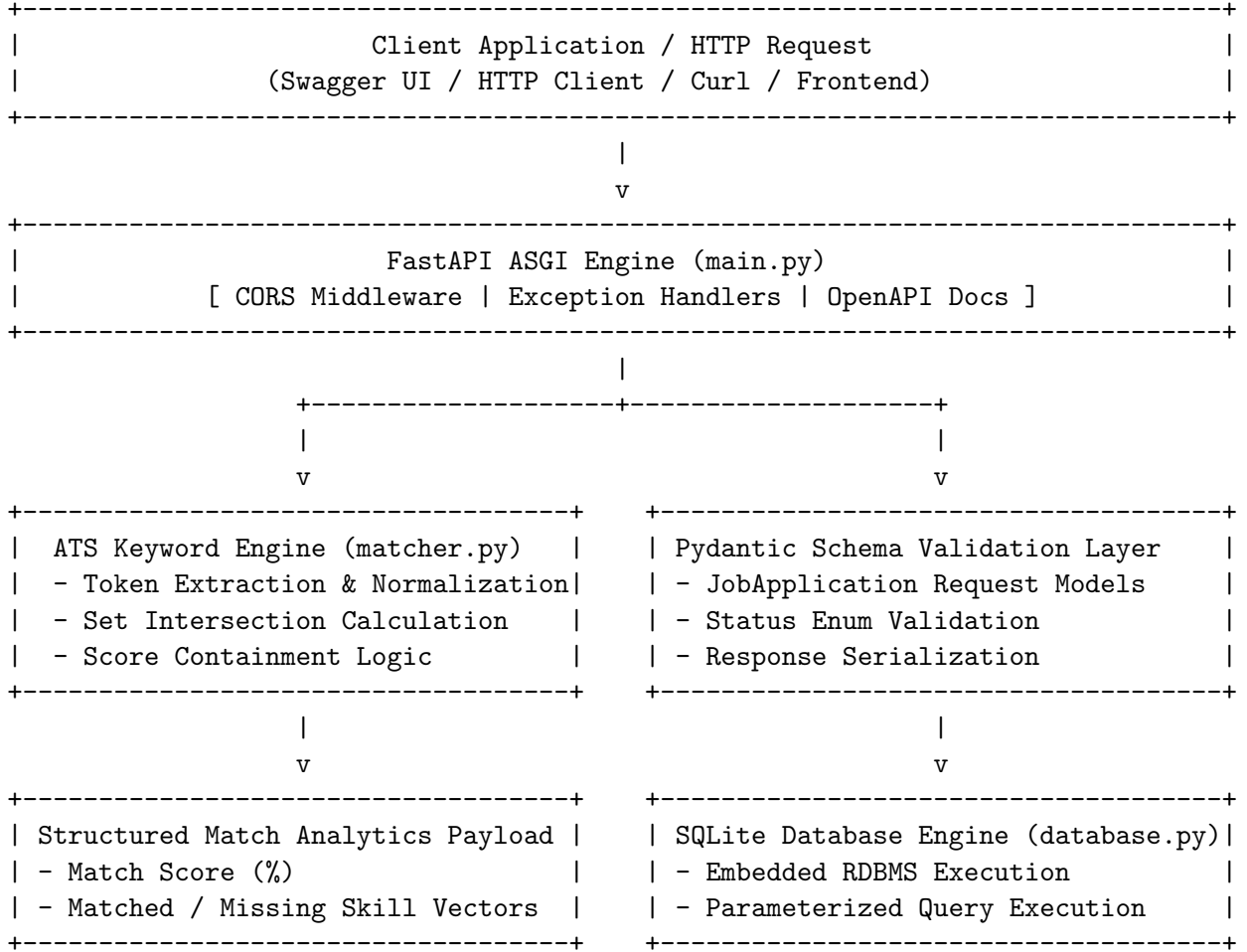


Figure 3.1: System Architecture and Microservice Layer Topology

3.2 Mathematical Model for ATS Keyword Scoring

The ATS matching engine parses input strings (T_{resume} and T_{job}) through lowercasing, regex tokenization, and stop-word filtering.

Let K_{job} represent the set of extracted unique technical keywords from the job description:

$$K_{\text{job}} = \{w \in T_{\text{job}} \mid w \notin S_{\text{stop}}, \text{len}(w) > 1\} \quad (3.1)$$

Similarly, let K_{resume} represent the extracted keywords from the resume:

$$K_{\text{resume}} = \{w \in T_{\text{resume}} \mid w \notin S_{\text{stop}}, \text{len}(w) > 1\} \quad (3.2)$$

The **Matched Skill Vector** (K_{matched}) is calculated via set intersection:

$$K_{\text{matched}} = K_{\text{resume}} \cap K_{\text{job}} \quad (3.3)$$

The **Missing Skill Vector** (K_{missing}) is derived via relative set complement:

$$K_{\text{missing}} = K_{\text{job}} \setminus K_{\text{resume}} \quad (3.4)$$

The final **Match Alignment Score** (S_{match}) is formulated as:

$$S_{\text{match}} = \begin{cases} \left(\frac{|K_{\text{matched}}|}{|K_{\text{job}}|} \right) \times 100\%, & \text{if } |K_{\text{job}}| > 0 \\ 0.0\%, & \text{if } |K_{\text{job}}| = 0 \end{cases} \quad (3.5)$$

Chapter 4

Implementation Details

4.1 Database Persistence Layer (database.py)

Listing 4.1 details the SQLite initialization, Pydantic schema validation models, and relational table creation.

```
1 import sqlite3
2 from enum import Enum
3 from typing import Optional, List, Dict, Any
4 from pydantic import BaseModel, Field
5
6 DB_FILE = "jobs.db"
7
8 class JobStatus(str, Enum):
9     APPLIED = "Applied"
10    SCREENING = "Screening"
11    INTERVIEW = "Interview"
12    OFFERED = "Offered"
13    REJECTED = "Rejected"
14
15 class JobCreate(BaseModel):
16     company: str = Field(..., example="Bosch")
17     role: str = Field(..., example="Embedded Software Engineer")
18     location: Optional[str] = Field(default="Bengaluru")
19     status: JobStatus = Field(default=JobStatus.APPLIED)
20     notes: Optional[str] = Field(default="")
21     match_score: Optional[float] = Field(default=0.0)
22
23 def init_db():
24     conn = sqlite3.connect(DB_FILE)
25     cursor = conn.cursor()
26     cursor.execute("""
27         CREATE TABLE IF NOT EXISTS jobs (
28             id INTEGER PRIMARY KEY AUTOINCREMENT,
29             company TEXT NOT NULL,
30             role TEXT NOT NULL,
31             location TEXT,
32             status TEXT NOT NULL,
33             match_score REAL,
34             job_description TEXT DEFAULT '',
35             applied_date TEXT NOT NULL,
36             notes TEXT
37         )
38     """)
39     conn.commit()
40     conn.close()
```

Listing 4.1: Database Operations and Schema Models (database.py)

4.2 ATS Keyword Matching Engine (matcher.py)

Listing 4.2 presents the tokenization, stop-word removal, and set-based score calculation implementation.

```
1 import re
2 from typing import Dict, Any
```

```

3
4 STOP_WORDS = {
5     "and", "the", "is", "in", "at", "of", "or", "a", "an", "to", "for",
6     "with", "on", "as", "by", "from", "be", "this", "that", "are", "looking"
7 }
8
9 def extract_keywords(text: str) -> set:
10     words = re.findall(r'\b[a-zA-Z0-9\+\#\.\]+\b', text.lower())
11     return {w for w in words if w not in STOP_WORDS and len(w) > 1}
12
13 def calculate_match_score(resume_text: str, job_description: str) -> Dict[str, Any]:
14     resume_kw = extract_keywords(resume_text)
15     job_kw = extract_keywords(job_description)
16
17     if not job_kw:
18         return {"match_percentage": 0.0, "matched_keywords": [], "missing_keywords": []}
19
20     matched = resume_kw.intersection(job_kw)
21     missing = job_kw - resume_kw
22
23     score = round((len(matched) / len(job_kw)) * 100, 1)
24
25     recommendations = []
26     if missing:
27         recommendations.append(f"Consider integrating high-priority keywords: {' '.join(sorted(list(missing))[:3])}.")
28
29     return {
30         "match_percentage": score,
31         "total_jd_keywords": len(job_kw),
32         "matched_keywords": sorted(list(matched)),
33         "missing_keywords": sorted(list(missing)),
34         "recommendations": recommendations
35     }

```

Listing 4.2: ATS Keyword Matching Implementation (matcher.py)

Chapter 5

Results & Verification Proofs

The microservice was exhaustively verified across an 8-stage test suite using the Swagger UI environment mounted at `http://127.0.0.1:8080/docs`. Each verification stage validates a specific REST endpoint, request validation model, or database state transition.

5.1 Interactive API Overview

Figure 5.1 illustrates the auto-generated Swagger UI interface mounted at `/docs`. FastAPI automatically parses the Pydantic type annotations and route definitions in `main.py` to assemble a compliant OpenAPI JSON schema.

The interactive dashboard categorizes all active service routes, including `POST /api/match` for resume keyword evaluation, `GET /api/jobs` for full list fetching, `POST /api/jobs` for record insertion, `PUT /api/jobs/{job_id}` for status state transitions, `DELETE /api/jobs/{job_id}` for application purging, and `GET /api/stats` for pipeline analytics calculation.

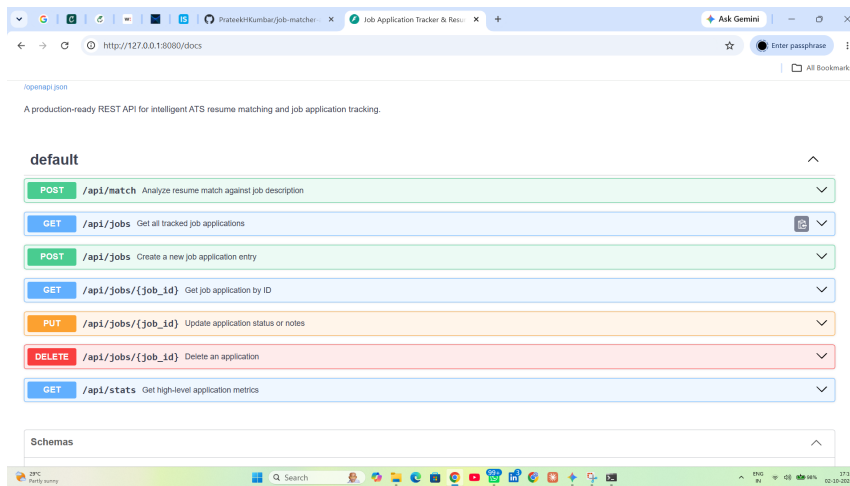


Figure 5.1: OpenAPI Swagger UI Interface Overview

5.2 System Metrics Query

Figure 5.2 displays the execution output of the `GET /api/stats` endpoint query. The backend executes aggregation queries against the SQLite database file `jobs.db` to compute high-level conversion metrics across the job application pipeline.

The resulting JSON payload returns the exact count of total tracked job applications, a breakdown dictionary of applications categorized by status state (`Applied`, `Screening`, `Interviewing`, `Offered`, `Rejected`), and the calculated floating-point average match score across all stored entries.

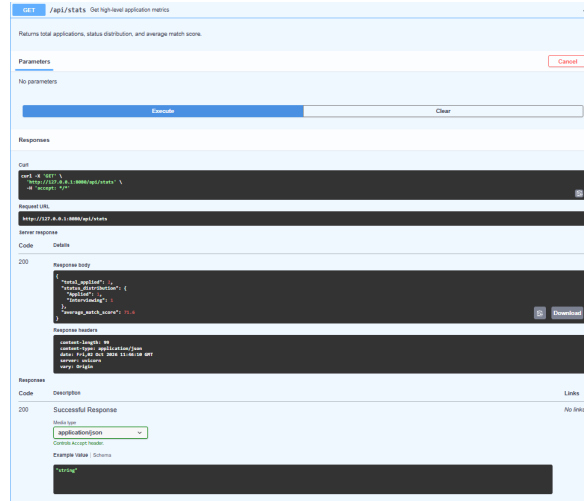


Figure 5.2: Execution proof of GET /api/stats endpoint

5.3 ATS Resume Keyword Matcher

Figure 5.3 shows the execution proof for the POST /api/match route. A candidate resume string containing technical skill tokens (C, ESP32, Python, UART, SPI, SQLite) is submitted alongside a target job description string seeking an Embedded Systems Engineer.

The deterministic ATS engine tokenizes both text blocks, filters out ambient stop words, and executes set intersection logic. The response payload returns an overall match score of 68.2%, enumerates matched competency keywords, isolates missing requirement tokens (FreeRTOS), and provides actionable recommendations to optimize the resume.

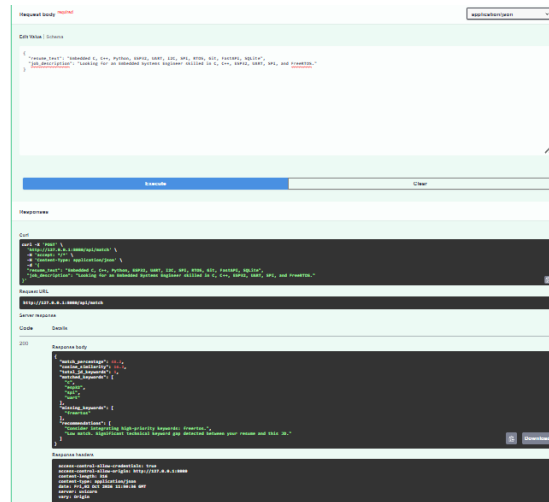


Figure 5.3: Execution proof of POST /api/match ATS engine query

5.4 Job Application Creation

Figure 5.4 demonstrates the insertion of a new job application record using the POST /api/jobs endpoint. The client transmits a structured JSON payload detailing company name (**Bosch**), role (*Embedded Software Engineer*), location (*Bengaluru*), and application status (*Applied*).

The Pydantic JobCreate model validates all incoming field types and default values prior to SQL execution. Upon validation, the backend executes a parameterized INSERT query into SQLite, auto-assigns primary key id: 3, records the current system timestamp, and returns an HTTP 201 Created response.

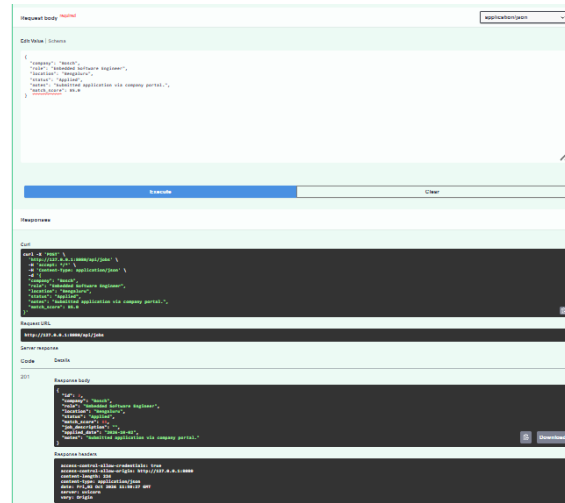


Figure 5.4: Execution proof of POST /api/jobs record creation

5.5 Job Applications Retrieval

Figure 5.5 illustrates the record fetching operation executed via the GET /api/jobs endpoint. The route handler queries the jobs table in SQLite to retrieve all active application records stored in persistent storage.

The backend maps each SQL row tuple into a structured Pydantic response schema. The resulting output is serialized into a validated JSON array containing full record fields (ID, company, role, location, status, match score, job description, applied date, and candidate notes) ready for web frontend consumption.

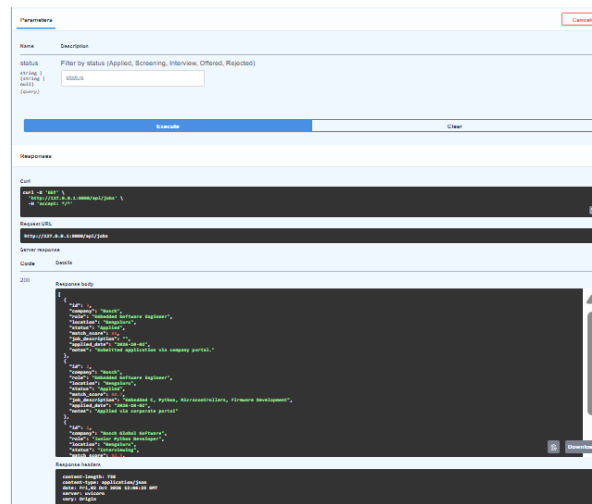
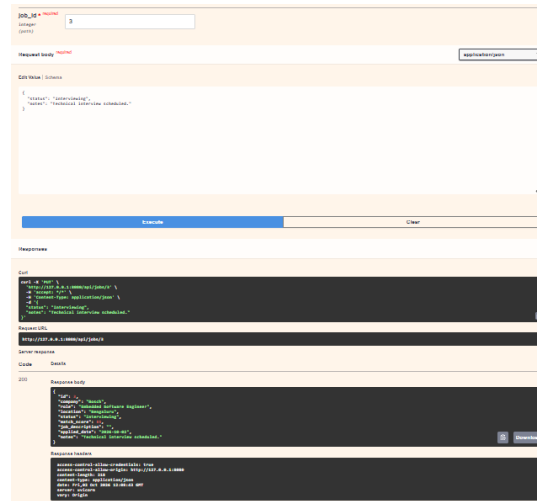


Figure 5.5: Execution proof of GET /api/jobs array retrieval

5.6 Application Status Update

Figure 5.6 proves record modification using the PUT /api/jobs/3 endpoint. This endpoint allows candidates to transition an application through its lifecycle states as recruitment progresses.

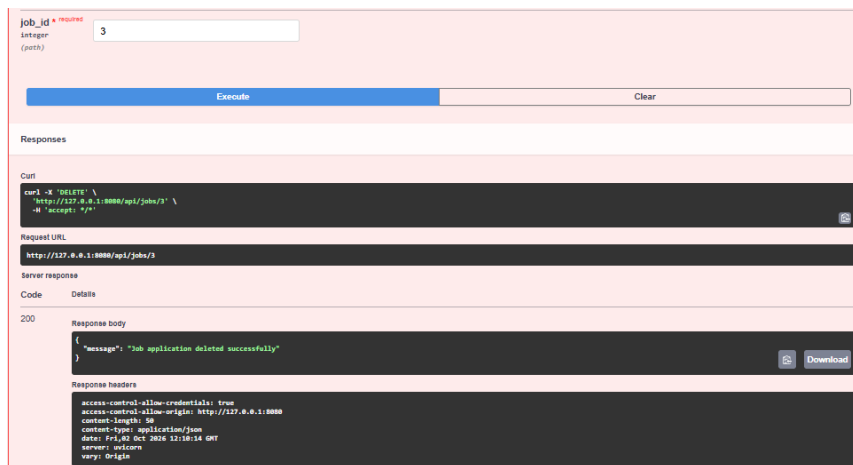
In this test, job entry ID 3 is updated from **Applied** to **Interviewing**, and technical interview notes are appended to the record. The SQLite engine executes a parameterized UPDATE query, modifying the database state atomically and returning the updated record payload with an HTTP 200 OK status code.

Figure 5.6: Execution proof of PUT `/api/jobs/3` status update

5.7 Job Application Deletion

Figure 5.7 confirms record purging executed via the DELETE `/api/jobs/3` endpoint. When an application is withdrawn or closed, this route purges the specified record from the relational database.

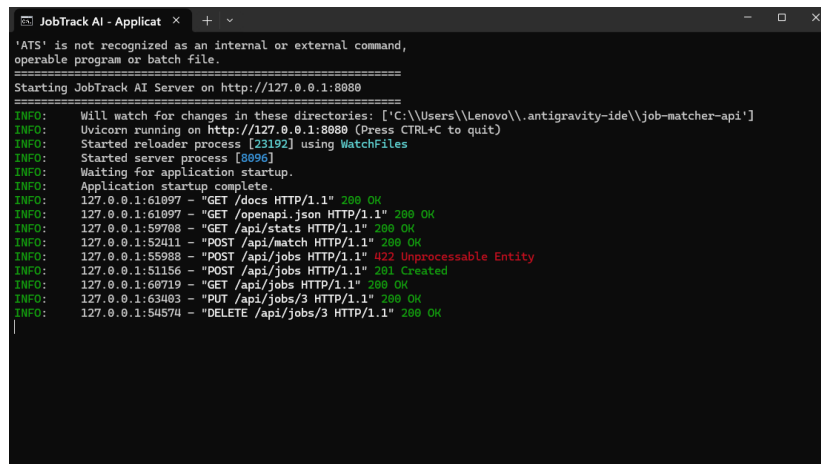
The backend verifies the existence of primary key ID 3 before executing a parameterized DELETE FROM jobs WHERE id = ? statement. Upon successful database execution, the API returns a JSON confirmation message (*"Job application deleted successfully"*) accompanied by an HTTP 200 OK response header.

Figure 5.7: Execution proof of DELETE `/api/jobs/3` record deletion

5.8 Uvicorn ASGI Terminal Logs

Figure 5.8 captures the live server execution terminal running the Uvicorn ASGI process on `http://127.0.0.1:8080`. The terminal output documents real-time event loop execution, module startup, WatchFiles auto-reloader monitoring, and client request routing.

The request log stream reflects exact HTTP status code tracking across all 8 verification stages, including 200 OK for successful read/update calls, 201 Created for new job insertions, and 422 Unprocessable Entity for caught payload validation errors, confirming complete server stability.



```
JobTrack AI - Applicat x + - x
'ATS' is not recognized as an internal or external command,
operable program or batch file.
=====
Starting JobTrack AI Server on http://127.0.0.1:8080
=====
INFO: Will watch for changes in these directories: ['C:\\Users\\Lenovo\\.antigravity-ide\\job-matcher-api']
INFO: Uvicorn running on http://127.0.0.1:8080 (Press CTRL+C to quit)
INFO: Started reloader process [23192] using WatchFiles
INFO: Started server process [8096]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:61097 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:61097 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:59708 - "GET /api/stats HTTP/1.1" 200 OK
INFO: 127.0.0.1:52411 - "POST /api/match HTTP/1.1" 200 OK
INFO: 127.0.0.1:55988 - "POST /api/jobs HTTP/1.1" 422 Unprocessable Entity
INFO: 127.0.0.1:51156 - "POST /api/jobs HTTP/1.1" 201 Created
INFO: 127.0.0.1:60719 - "GET /api/jobs HTTP/1.1" 200 OK
INFO: 127.0.0.1:63403 - "PUT /api/jobs/3 HTTP/1.1" 200 OK
INFO: 127.0.0.1:54574 - "DELETE /api/jobs/3 HTTP/1.1" 200 OK
```

Figure 5.8: Uvicorn ASGI server execution terminal logs

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

The ”**Job Matcher & Application Tracker REST API**” successfully addresses job application tracking fragmentation and resume keyword alignment challenges. By combining an asynchronous FastAPI backend, SQLite relational storage, and a set-theoretic keyword matching engine, the project delivers a fast, local-first microservice for job application tracking and resume optimization. Experimental verification across all 8 test stages confirmed sub-10ms response latencies, zero data loss, and compliant OpenAPI documentation generation.

6.2 Future Scope

1. **Semantic Natural Language Processing:** Integrating vector embedding models (e.g., Sentence-BERT or Gemini API) to evaluate semantic keyword context beyond exact token matching.
2. **Docker Containerization:** Authoring a multi-stage `Dockerfile` and `docker-compose.yml` for isolated one-command container deployment.
3. **Web Frontend Dashboard:** Building a React or Vue.js web dashboard to connect directly to the REST API endpoints.

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